

NELSON LAGOON, AK, USA

WMO: 703058

Lat: 56.010N

Lon: 161.160W

Elev: 14

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 00110

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	8.5	10.1	1.0	5.8	7.7	5.0	7.1	10.8	39.4	28.8	35.6	27.3	14.4	330	0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	61.0	57.3	58.9	56.1	57.2	54.8	58.1	60.1	56.6	58.5	55.2	56.9	13.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	69.3	58.8	55.4	65.3	57.3	54.3	62.8	56.4	25.1	60.1	24.1	58.6	23.4	57.0	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.9	32.4	28.5	4.4	66.6	8.2	3.2	-1.4	68.8	-6.2	70.7	-10.8	72.5	-16.7	74.8		
(5)			4.9	63.0	9.4	3.8	-1.8	65.7	-7.4	68.0	-12.6	70.1	-19.5	72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	28.0	29.9	27.8	33.6	40.1	46.2	51.0	53.1	49.9	42.9	36.0	32.4		
	DBStd	10.55	8.31	7.35	8.64	5.79	4.93	3.95	3.77	3.34	3.61	4.52	5.29	6.06		
	HDD50	4130	683	563	687	493	308	125	33	6	43	223	419	547		
	HDD65	9385	1148	983	1152	943	773	564	434	369	453	685	869	1012		
	CDD50	219	0	0	0	0	0	10	64	101	40	4	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	13.2	15.2	13.3	14.3	14.3	10.8	10.8	9.8	10.9	14.5	13.9	14.5	16.3		
	PrecAvg	41.4	3.9	3.5	2.8	2.3	2.6	3.0	3.7	3.0	4.4	3.8	4.0	4.4		
(15) Precipitation	PrecMax	56.3	6.3	7.3	5.8	3.7	4.9	4.9	5.7	4.4	7.2	6.5	6.6	9.4		
	PrecMin	31.1	1.6	1.1	0.8	0.7	0.7	1.3	1.5	1.1	3.0	1.9	1.5	1.6		
	PrecStd	6.1	1.3	1.5	1.3	0.8	1.0	0.9	1.1	0.8	1.1	1.2	1.3	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	43.3	44.6	47.6	52.3	58.9	63.3	65.1	61.9	54.7	47.4	45.9		
		MCWB	40.0	40.6	41.8	43.9	47.8	54.9	58.3	60.6	58.3	51.3	45.1	44.0		
	2%	DB	40.1	41.2	41.1	44.8	49.6	55.3	60.3	61.3	58.5	52.9	45.9	43.2		
		MCWB	38.3	39.0	39.1	41.5	46.1	52.1	56.3	57.9	55.4	49.9	44.1	41.1		
	5%	DB	38.7	39.6	39.1	42.8	47.9	53.3	57.9	59.4	56.5	51.5	44.5	41.2		
		MCWB	37.1	37.7	37.2	40.0	45.1	50.5	54.9	56.9	53.6	49.0	42.7	39.3		
	10%	DB	37.1	38.1	37.5	40.8	46.3	51.5	56.2	57.8	54.8	49.4	43.0	39.3		
		MCWB	35.8	36.3	35.8	38.7	43.7	49.1	53.8	55.6	52.1	46.9	41.2	37.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	41.5	42.1	44.7	50.2	55.5	59.2	61.6	58.8	52.2	46.2	44.3		
		MCDB	41.3	42.9	44.0	47.5	51.6	57.7	62.0	62.8	61.2	53.6	46.6	45.3		
	2%	WB	38.8	39.5	39.4	41.9	47.3	52.8	57.0	58.7	56.2	50.5	44.6	41.6		
		MCDB	39.8	40.7	40.6	44.1	48.6	54.6	59.4	60.6	58.1	52.3	45.4	42.8		
	5%	WB	37.4	37.9	37.4	40.4	45.9	51.0	55.5	57.3	54.1	49.1	43.1	39.6		
		MCDB	38.4	39.2	38.5	42.4	47.4	52.8	57.3	59.0	55.9	51.2	44.3	41.0		
	10%	WB	36.1	36.6	36.2	38.9	44.4	49.3	54.3	56.0	52.5	47.0	41.4	38.0		
		MCDB	37.2	37.9	37.4	40.7	46.1	51.0	55.9	57.6	54.3	48.9	43.0	39.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.0	7.5	6.6	6.5	6.1	6.2	5.7	5.3	5.9	6.2	6.1		
		MCDBR	6.3	7.6	7.2	8.1	8.5	9.1	9.3	8.0	7.1	6.6	6.5	6.4		
		MCWBR	5.5	6.5	6.0	6.1	6.0	6.3	6.4	6.1	5.7	5.3	6.1	5.7		
	5% WB	MCDBR	6.1	7.3	7.2	7.6	7.9	8.7	8.7	8.0	6.9	6.3	6.3	6.2		
		MCWBR	5.7	6.6	6.2	6.1	6.2	6.5	6.4	6.1	6.0	5.6	6.4	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.327	0.340	0.387	0.390	0.385	0.377	0.389	0.364	0.340	0.322	0.280		
	taud		2.308	2.293	2.270	2.183	2.269	2.372	2.451	2.389	2.399	2.325	2.173	2.321		
	Ebn at Noon		199	232	260	262	269	272	273	260	250	223	171	173		
	Edh at Noon		16	24	31	40	39	36	32	32	28	23	18	13		
(44) All-Sky Solar Radiation	RadAvg		149	318	743	1166	1422	1507	1305	1069	773	439	188	103		
	RadStd		19	36	121	89	133	156	136	98	71	34	18	10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	+2.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page